IS 525: Final Project Report
The Superstore Retail Analytics Dashboard

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Abstract

The **Superstore Retail Analytics Dashboard** project focuses on transforming transactional retail data into actionable business intelligence using **Power BI**. The project analyzes sales, profitability, geography, product portfolio performance, and operational factors using the publicly available *Superstore* dataset.

Through structured data cleaning, modeling, and the creation of calculated measures (such as total sales, total profit, profit margins, year-over-year growth, and delivery performance metrics), the dashboard enables stakeholders to explore performance dynamically rather than relying on static reports.

The dashboards address key business questions related to **which products and regions drive profitability**, how sales are distributed across categories and sub-categories, and how operational decisions—such as shipping modes and delivery speed—impact returns and margins. Interactive filters allow users to drill down by category, region, and customer behavior, enabling multi-dimensional analysis.

Overall, this project demonstrates how modern BI tools can convert raw transactional data into strategic insights, improving operational transparency and supporting data-driven decision-making in retail environments.

Problem Statement

Retail organizations often generate large volumes of transactional data, but **identifying meaningful insights from this data remains challenging**. Sales volume alone does not reflect business success; high revenue products or regions may still generate low or negative profits due to operational inefficiencies, discounts, or returns.

Key challenges addressed in this project include:

- Difficulty identifying **profitable vs underperforming product sub-categories**
- Limited visibility into **regional contribution to total company profit**
- Inability to easily assess how **shipping modes and delivery speed affect returns and margins**
- Fragmented reporting that prevents stakeholders from viewing performance holistically

Without an integrated and interactive analytics solution, decision-makers risk allocating resources inefficiently and overlooking critical risk areas within the business.

Purpose Statement

The purpose of this project is to design an **interactive Power BI dashboard** that supports strategic and operational decision-making in a retail context. Specifically, the dashboard aims to:

1. Identify **high-performing and underperforming product portfolios** using sales-profit quadrant analysis
2. Analyze **regional and state-level profit concentration** to highlight growth opportunities
3. Evaluate **operational drivers**, including shipping methods and delivery timelines, that influence profitability and return behavior
4. Enable stakeholders to explore trends and relationships through intuitive visualizations and filters

By presenting complex data in an accessible visual format, the dashboard empowers stakeholders to make informed, evidence-based decisions.

Intended Audience

The dashboard is designed for **non-technical and technical stakeholders** involved in retail strategy and operations, including:

- **Executive leadership** – to understand overall business performance and profit drivers
- **Sales and category managers** – to optimize product mix and pricing strategies
- **Regional managers** – to assess geographic performance and regional contributions
- **Operations and logistics teams** – to evaluate shipping efficiency and delivery impact
- **Finance and strategy teams** – to monitor profitability, margins, and risk areas

The design prioritizes clarity and usability so insights can be understood without advanced analytical training.

Data Source

- **Dataset:** Superstore Dataset
- **Source:** Kaggle (publicly available)
- **Link:** https://www.kaggle.com/datasets/vivek468/superstore-dataset-final/data

The dataset contains retail transactional records including order details, product information, customer segments, geographic attributes, sales, profit, and shipping data. This dataset is commonly used for retail analytics and is suitable for demonstrating BI concepts and techniques.

Data Analysis

The analysis focuses on multiple dimensions of retail performance:

- **Sales Trends:** Monthly and year-over-year sales patterns to identify growth and seasonality
- **Product Portfolio:** Category and sub-category performance using sales, profit, and profitability quadrants
- **Geographic Performance:** Region and state-level contribution to total profit and sales concentration
- **Operational Efficiency:** Analysis of shipping modes, delivery days, and their relationship to returns and margins
- **Customer & Payment Behavior:** Evaluation of customer segments and payment methods (where applicable) to identify behavioral patterns

Each dashboard emphasizes a specific analytical perspective while remaining interconnected through shared filters and measures.

Methods

Data Preparation

- Cleaned the dataset using **Power Query**
- Removed duplicates and handled missing values
- Standardized categorical fields (categories, regions, ship modes)
- Created derived fields such as **Delivery Days** (Ship Date − Order Date)

Data Modeling

- Established appropriate relationships between tables
- Ensured correct granularity for time-based and geographic analysis
- Verified data integrity for aggregated measures

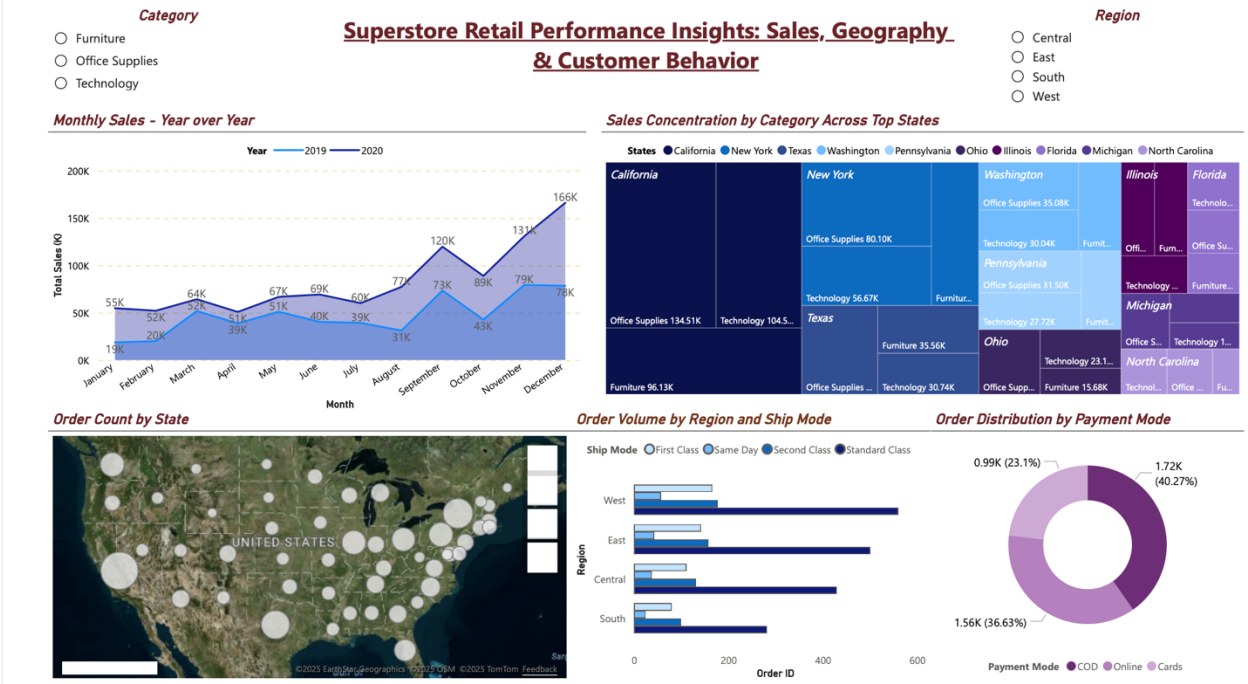
DAX Measures

- Created measures for **Total Sales**, **Total Profit**, **Profit Margin %**, **Year-over-Year Sales**, **Average Delivery Days**, **Return Rate**, and **normalized KPIs**
- Developed ranking and contribution measures for comparative analysis

Visualization Techniques

- KPI cards for high-level metrics
- Line charts for trend analysis
- Maps and treemaps for geographic distribution
- Scatter/bubble charts for profitability quadrants
- Waterfall charts for regional profit contribution
- Sankey and chord diagrams (custom visuals) for inter-category and product flow relationships

Dashboard 1: Sales, Geography & Customer Behavior Overview



Dashboard Objective

The objective of Dashboard 1 is to provide a **high-level overview of retail sales performance**, focusing on **temporal trends, geographic distribution, customer purchasing behavior, and payment preferences**. This dashboard is designed to help stakeholders quickly understand **where sales are coming from, how they evolve over time, and how customers interact with the business across regions and channels**.

Stakeholders

This dashboard is primarily intended for:

- **Business executives and senior management** seeking an overall performance snapshot
- **Sales and marketing teams** evaluating regional demand and category performance
- **Operations managers** monitoring shipping patterns and order volumes
- **Financial analysts** assessing revenue concentration and payment behavior

Visualizations Explained

1. Monthly Sales – Year over Year

This line and area chart compares **monthly total sales for 2019 and 2020**, allowing users to identify seasonal trends and growth patterns. The visualization highlights **sales acceleration in the latter half of the year**, particularly during Q4, indicating strong year-end demand.

2. Sales Concentration by Category Across Top States
This treemap displays how **sales are distributed across top-performing states**, broken down by **product category**. Larger blocks represent higher sales contributions, making it easy to identify **states and categories that dominate revenue generation**, such as California and New York.

3. Order Count by State (Geographic Map)
The map visualization plots **order volume geographically across the United States**, with bubble size representing the number of orders. This provides spatial context and helps identify **regional demand clusters**, highlighting coastal and high-population states as major order hubs.

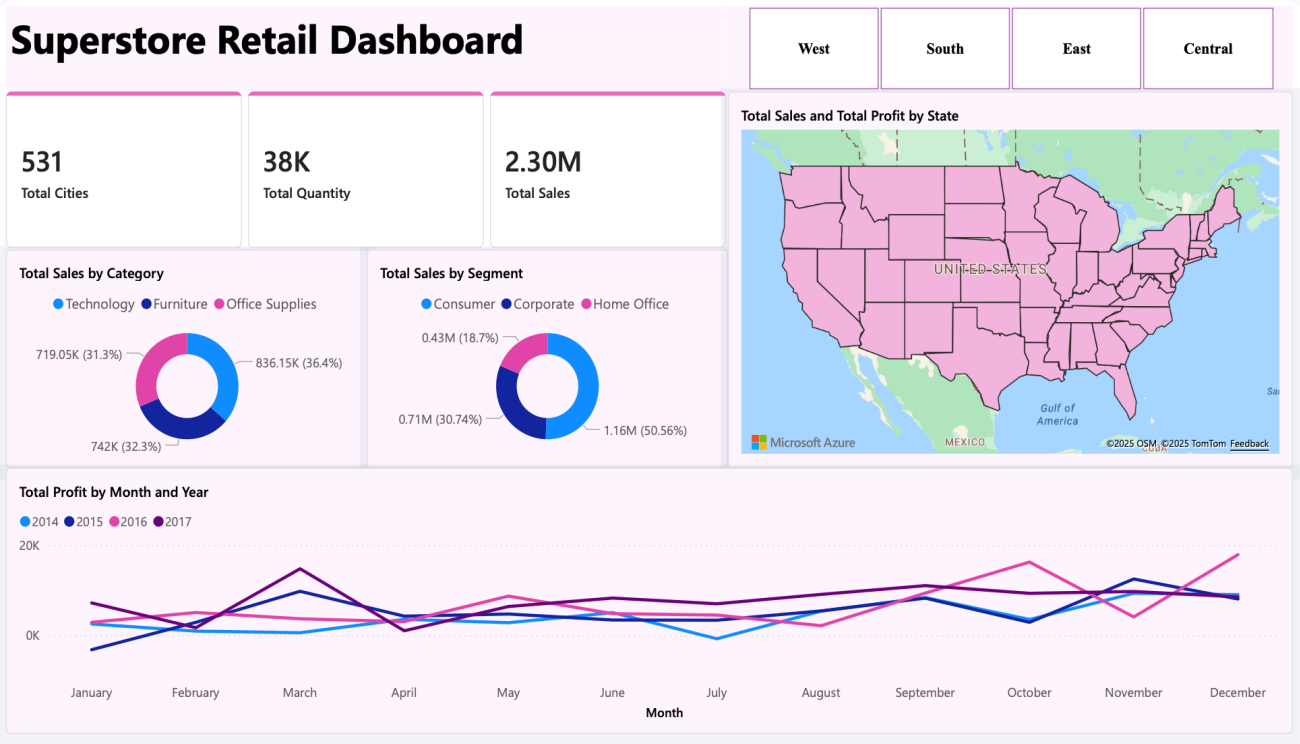
4. Order Volume by Region and Ship Mode
This horizontal bar chart compares **order volumes across regions (Central, East, South, West)** by **shipping mode**. It allows stakeholders to understand customer preferences for shipping options and assess how logistics strategies vary by region.

5. Order Distribution by Payment Mode
This donut chart shows the **proportion of orders by payment method**, such as COD, online, and card payments. It provides insight into **customer payment behavior**, which is useful for evaluating transaction costs, fraud risk, and operational efficiency.

Insights Gained

- Sales exhibit **strong seasonality**, with significant growth during the final quarter of the year.
- A **small number of states contribute a disproportionate share of total sales**, indicating geographic concentration.
- Coastal regions, particularly the **West and East**, generate higher order volumes than Central and South regions.
- **Standard Class shipping** dominates order volume across most regions, suggesting cost sensitivity among customers.
- **Online and card payments account for the majority of transactions**, reducing reliance on cash-based methods and supporting digital-first operations.

Dashboard 2: Overall Retail Performance Overview



Dashboard Objective

The objective of this dashboard is to provide a **comprehensive, high-level overview of Superstore's retail performance** across products, customers, geography, and time. While other dashboards dive into specific strategic or operational questions, this dashboard acts as the **primary entry point** for understanding the business as a whole.
It is designed to answer foundational questions such as:

- How large is the business in terms of sales volume and geographic reach?
- Which product categories and customer segments contribute the most to revenue?
- How does profitability change across months and years?
- Are there visible geographic patterns that influence sales and profit performance?

By consolidating key metrics and trends into a single view, this dashboard enables stakeholders to quickly assess overall performance before drilling down into more detailed analyses.

Stakeholders

- **Executive Leadership & Senior Management**
  This dashboard supports high-level decision-making by summarizing overall business performance, revenue scale, and profitability trends in a single view.
- **Category Managers & Marketing Teams**
  prioritize promotional strategies and product investments.
- **Operations & Supply Chain Teams**

The geographic distribution of sales across states and cities helps these teams plan inventory allocation, shipping routes, and regional logistics more effectively.

- **Business Analysts & Strategy Team**
  The dashboard serves as a baseline reference for identifying areas that require deeper investigation using more specialized dashboards.

Visual Explanation

This dashboard is intentionally structured to move from **summary → distribution → trend → geography**, creating a clear analytical narrative:

- **KPI Cards (Top Section)**
  Highlight Total Cities (531), Total Quantity sold, and Total Sales (2.30M). These metrics immediately communicate the scale and reach of the business.
- **Donut Charts – Sales Distribution**
  - *Total Sales by Category* shows how revenue is split across Technology, Furniture, and Office Supplies.
  - *Total Sales by Segment* compares Consumer, Corporate, and Home Office customers, helping identify the most valuable customer base.
- **Filled Map – Total Sales & Profit by State**
  Aggregates performance at the state level to clearly display geographic concentration and avoid clutter from city-level plotting.
- **Line Chart – Total Profit by Month and Year**
  Displays month-by-month profit trends across multiple years, making seasonality and long-term performance changes easy to identify.
- **Bar Chart – Order Volume by Region and Ship Mode**
  Compares shipping patterns across regions, revealing operational differences in how orders are fulfilled.
- **Donut Chart – Order Distribution by Payment Mode**
  Shows customer payment preferences across COD, Online, and Card payments, providing insight into transaction behavior.

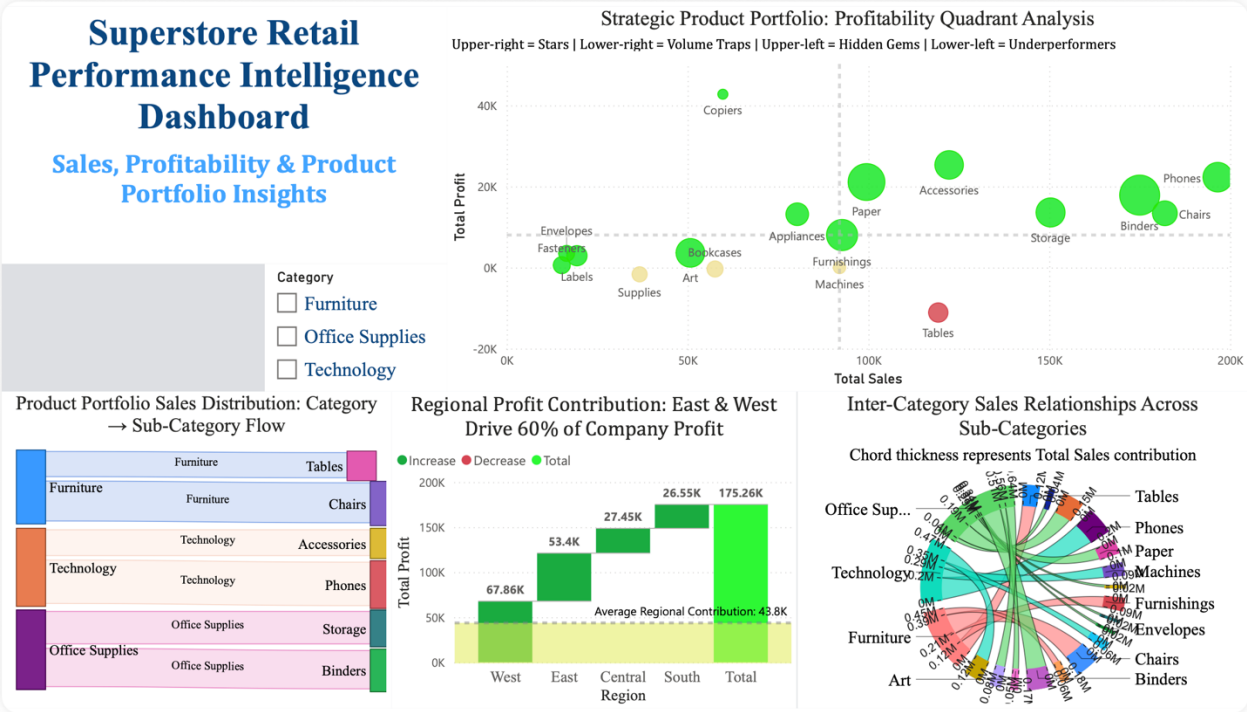
Insights Gained

- **Balanced Product Portfolio**
  Revenue is evenly distributed across categories, with Technology (36.4%), Furniture (32.3%), and Office Supplies (31.3%) contributing similar shares. This balance reduces business risk associated with over-reliance on a single product line.
- **Consumer Segment Is the Primary Revenue Driver**
  The Consumer segment contributes more than half of total sales (50.56%), significantly outperforming Corporate and Home Office segments. This indicates that strategies targeting individual consumers are critical for revenue growth.
- **Wide Geographic Footprint Increases Operational Complexity**
  Operating across 531 cities highlights the scale of logistics and the importance of geographic insights for supply chain planning.
- **Clear Seasonal Profit Patterns**
  Monthly profit trends reveal recurring patterns:
  - February and April often show weaker performance
  - End-of-year months consistently generate higher profits

    - Overall profit trends show improvement over time, particularly in later years
- **Strong Regional Performance Differences**
    - The West region leads in overall sales, driven largely by states like California and Washington
    - The East region performs strongly due to high-profit states such as New York
    - Central shows steady but lower-volume performance
    - The South displays mixed results, with some states showing strong sales but weaker profitability

These insights help stakeholders identify where to focus growth strategies, cost controls, and regional optimization efforts.

Dashboard 3: Strategic Product Portfolio & Regional Profitability Analysis



Objective

The objective of this dashboard is to **analyze product-level performance and regional profitability from a strategic perspective**, moving beyond simple sales totals to identify **which products truly drive value** and which require intervention.

This dashboard is designed to answer higher-level business questions such as:

- Which sub-categories are high-performing "stars" versus underperformers?
- Are there products generating high sales but weak profitability?
- How do different regions contribute incrementally to total company profit?
- How are sales distributed across categories and sub-categories?

By combining **portfolio analysis, regional contribution, and flow-based visuals**, this dashboard supports **strategic decision-making** related to product investment, discontinuation, and regional prioritization.

Stakeholders

- **Executive Leadership & Strategy Teams**
  Use this dashboard to evaluate product portfolio health, identify strategic growth opportunities, and decide where to allocate resources.
- **Product & Category Managers**
  Can quickly identify "star" products to scale, "hidden gems" to nurture, and underperforming sub-categories that may need cost reduction, repositioning, or removal.
- **Regional Sales & Operations Managers**

Benefit from understanding how each region contributes to total profit and where regional performance deviates from the average.

- **Business Analysts**
  Use this dashboard as a strategic diagnostic tool before conducting deeper operational or customer-level analysis.

Visual Explanation

This dashboard combines **three advanced analytical perspectives**:

1. Profitability Quadrant (Bubble Scatter Plot)

- **X-axis:** Total Sales
- **Y-axis:** Total Profit
- **Bubble Size:** Sales volume
- **Bubble Color:** Profitability status

Dashed reference lines divide the chart into four strategic quadrants:

- **Upper-right (Stars):** High sales, high profit (e.g., Phones, Chairs)
- **Upper-left (Hidden Gems):** Low sales, high profit (e.g., Copiers)
- **Lower-right (Volume Traps):** High sales, low/negative profit (e.g., Tables)
- **Lower-left (Underperformers):** Low sales, low profit

This visualization enables fast identification of **products that deserve expansion versus scrutiny**.

2. Regional Profit Contribution Waterfall Chart

This chart decomposes **total company profit** into regional contributions:

- Starts with individual region profit (West, East, Central, South)
- Ends with Total Company Profit
- Includes an **average regional contribution reference line**

Color coding highlights regions that:

- Contribute **above average** (positive drivers)
- Contribute **below average** (potential improvement areas)

This helps stakeholders clearly see **which regions are pulling their weight** and which are not.

3. Category → Sub-Category Flow (Sankey Diagram)

The Sankey diagram visualizes how **total sales flow from categories to sub-categories**:

- Left side: Categories (Furniture, Technology, Office Supplies)
- Right side: Sub-categories (e.g., Chairs, Phones, Storage, Binders)
- Flow thickness represents **sales contribution**

This visual makes it easy to:

- Identify dominant sub-categories within each category
- Spot over-reliance on specific products
- Understand portfolio concentration at a glance

4. Inter-Category Sales Relationships (Chord Diagram)

The chord diagram shows **sales relationships across categories and sub-categories**:

- Each arc represents a category or sub-category
- Chord thickness represents total sales volume
- Color coding distinguishes category groupings

This advanced visualization highlights:

- Strong interconnections between high-performing categories
- How sub-categories collectively contribute to overall sales
- Portfolio balance and diversification patterns

Insights Gained

- **Clear Product "Stars" and "Volume Traps" Exist**
  Products like *Phones* and *Chairs* generate both high sales and profit, making them ideal candidates for further investment. In contrast, *Tables* show high sales but negative profit, signaling pricing or cost issues.
- **Hidden Profit Opportunities Identified**
  Sub-categories such as *Copiers* deliver strong profits despite lower sales volumes, indicating untapped growth potential.
- **Regional Profit Is Unevenly Distributed**
  The West and East regions together drive approximately **60% of total company profit**, while Central and South contribute less than the regional average.
- **Sales Are Concentrated Within Key Sub-Categories**
  A small number of sub-categories account for a disproportionately large share of total sales, increasing dependency risk if demand shifts.
- **Strategic Portfolio Balance Is Required**
  The combination of strong performers, hidden gems, and underperformers highlights the need for **targeted strategies**, rather than one-size-fits-all decisions across products and regions.

Key Insights, Challenges, Adjustments & Feedback

Key Insights (Across All Dashboards)

Across the three dashboards, several important business insights emerged when analyzing the Superstore dataset from multiple perspectives—overall performance, operational patterns, and strategic portfolio positioning.

1. **Revenue and Profit Are Not Always Aligned**
   While some categories and sub-categories generate high sales volumes, they do not always translate into high profitability. This was especially evident in product portfolio analysis, where certain high-volume items acted as "volume traps" with weak or negative profit margins.
2. **Consumer Segment Drives the Majority of Sales**
   The Consumer segment consistently emerged as the largest contributor to total sales, outperforming Corporate and Home Office segments. This insight suggests that customer-focused strategies targeting consumer behavior can have a significant impact on overall performance.
3. **Regional Performance Is Highly Concentrated**
   Sales and profit are not evenly distributed across regions. The West and East regions together account for a disproportionate share of total company profit, while other regions contribute below the average. This highlights opportunities for targeted regional strategies rather than uniform national approaches.
4. **Product Portfolio Has Clear Strategic Categories**
   The profitability quadrant analysis revealed distinct groups of products:
   - High-sales, high-profit "stars" suitable for scaling
   - High-profit but low-sales "hidden gems" with growth potential
   - High-sales but low-profit items requiring cost or pricing review
   - Underperformers that may need repositioning or discontinuation
5. **Seasonality and Timing Matter**
   Monthly trend analysis showed recurring seasonal patterns, including end-of-year profit spikes and consistent dips in certain months. These trends are critical for forecasting, inventory planning, and promotional timing.

Challenges & Resolutions

1. **Dashboard Overcrowding**
   Initially, too many visuals and metrics were included, which made the dashboards difficult to read and overwhelming for non-technical users.
   Resolution:
   Visuals were reduced to those with the highest business value, and dashboards were restructured to follow a clear narrative flow rather than a dense grid layout.
2. **Geographic Data Clutter**
   Mapping data at the city level resulted in excessive overlap and poor readability due to the large number of cities.

Resolution:
Geographic data was aggregated to the state level, improving clarity while still preserving meaningful geographic insights.

3. **Comparability Across Metrics**
   Metrics such as sales, profit, return rate, and delivery speed existed on different scales, making direct comparison difficult.
   Resolution:
   Normalized and score-based measures were introduced, allowing cross-metric comparison in advanced visuals such as radar and quadrant charts.
4. **Complex Visual Interpretation**
   Advanced visuals (e.g., Sankey and chord diagrams) required careful labeling and explanation to ensure they were interpretable by non-expert users.
   Resolution:
   Titles, subtitles, legends, and explanatory text were refined to clearly communicate what each visual represents and how it should be read.

Deviations from the Original Plan

Several adjustments were made as the project evolved:

- Some initially planned visuals (e.g., highly detailed sub-category and shipping analyses) were removed to reduce clutter.
- The dashboard design shifted from a darker, more stylized theme to a lighter, more academic and presentation-friendly format.
- Geographic analysis was simplified from city-level to state-level aggregation.
- The final project emphasized **strategic storytelling** over exhaustive metric coverage, ensuring dashboards remained accessible to non-technical stakeholders.

These deviations improved clarity, usability, and alignment with the project's analytical goals.

External Feedback & Incorporation

Feedback was obtained from individuals outside the course (including a roommate and peers), who reviewed the dashboards from a non-technical perspective. Key feedback included:

- Improve spacing and visual alignment
- Reduce visual clutter
- Maintain consistent color schemes
- Increase font readability
- Avoid overly complex visuals for first-time users

Actions Taken in Response:

- Reorganized dashboard layouts for cleaner alignment and flow
- Removed redundant or low-value visuals
- Standardized color palettes across dashboards
- Increased font sizes and label clarity
- Simplified explanations for advanced visualizations

Incorporating this feedback significantly improved the dashboards' clarity, professionalism, and usability for a broader audience.